

CADEMAS-ML: A Software Tool for Context-Aware Cooperative and Explainable Automated Decision-Making

Pavel Novoa-Hernández^{1,*} , Mariia Godz²  and David A. Pelta² 

¹ Dept. Computer Engineering and Systems, Universidad de La Laguna; pnovoah@ull.edu.es

² Dept. Computer Science and Artificial Intelligence, Universidad de Granada; {mariagodz,dpelta}@ugr.es

* Correspondence: pnovoah@ull.edu.es

Abstract

Automated decision-making systems increasingly operate as part of larger decision ecosystems rather than as isolated models. However, their outputs are often interpreted without considering the organizational, operational, or strategic context in which decisions are made. This paper presents CADEMAs-ML, a software tool that operationalizes the CADEMAs framework for a machine-learning-based class of cooperative and context-aware automated decision-making systems. While CADEMAs is a general conceptual framework for coordinating heterogeneous automated decision units, CADEMAs-ML specializes this idea to settings in which the contributing units are predictive models and the operational goal is case prioritization. The tool assigns weights to predictive components according to selected performance indicators and aggregates their outputs into a global risk score. In parallel, it evaluates contextual suitability using configurable fuzzy membership functions, categorical mappings, derived rules, and aggregation operators. The final prioritization score combines the predictive risk component and the contextual suitability component through an adjustable trade-off parameter. Beyond the original CADEMAs framework and the employee-attribution instantiation of Godz et al. [1], CADEMAs-ML introduces two operational modules: a robustness analysis module that evaluates ranking stability under alternative trade-off policies, and an Explain module that produces case-level decompositions, local perturbation-based ML attributions, fuzzy-rule traceability, and natural-language recommendations. The capabilities of the tool are illustrated through an employee attrition case study that instantiates, rather than redefines, CADEMAs in an ML workflow. Several departmental predictive models are combined with alternative decision contexts, including economic crisis and digital transformation scenarios. The results show that identical predictive risks may lead to different prioritization outcomes depending on the contextual assumptions adopted, and that individual cases can be audited through both quantitative traces and textual explanations. CADEMAs-ML therefore provides a practical bridge between cooperative ADM theory, fuzzy context modeling, explainable prioritization, and auditable decision-support software.

Keywords: automated decision-making systems; fuzzy decision making; context-aware decision support; machine learning; prioritization

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1. Introduction

Automated decision-making (ADM) systems have become an integral part of modern organizations. Following the broad understanding of ADM as computational processes that produce ratings, recommendations, decisions, predictions, or related outputs from data and

predefined objectives [2], such systems may include machine-learning classifiers, rule-based procedures, optimization models, simulation components, or other algorithmic decision units. In practice, predictive and classification models are among the most common instantiations, supporting tasks such as credit scoring, fraud detection, medical triage, customer churn analysis, and human-resource management.

Despite these advances, an important limitation remains. The outputs generated by ADM units, such as recommendations, ratings, class probabilities, or risk scores, are often interpreted in isolation from the organizational environment in which decisions are ultimately made [3,4]. In practice, a technically plausible automated output rarely provides sufficient guidance for action. The appropriate response depends on contextual factors such as organizational priorities, resource constraints, departmental objectives, regulatory constraints, and prevailing strategic conditions. As a result, identical automated assessments may lead to different decisions under different circumstances.

This observation highlights the cooperative nature of real-world decision-making [3]. Decisions are seldom based on a single decision agent or automated component. Organizations frequently rely on heterogeneous units developed for different departments, objectives, data sources, or stakeholder groups. Their outputs must therefore be harmonized into a coherent assessment before they can support action. At the same time, domain experts contribute contextual knowledge that cannot always be represented through precise numerical rules. Such knowledge is often expressed through gradual concepts and linguistic assessments that reflect uncertainty, preference, and strategic intent.

Fuzzy logic provides a well-established mathematical framework for representing this type of contextual reasoning [5,6]. Through fuzzy sets and linguistic variables, experts can model concepts such as urgency, strategic relevance, or suitability in a transparent and interpretable manner. This makes fuzzy approaches particularly attractive for decision-support environments in which quantitative predictions must be complemented by qualitative contextual considerations.

To address these challenges, Novoa-Hernández et al. [7] proposed **CADEMAS** (Cooperative Automated Decision-Making Systems), a conceptual framework for coordinating heterogeneous automated decision units. CADEMAs separates the decision process into an input-management layer, a set of ADM units, an output or decision-integration layer, and a contextual component that evaluates the integrated proposal against external conditions. Importantly, the framework is not restricted to predictive models. A CADEMAs unit may produce a score, recommendation, rule activation, symbolic output, or preliminary decision signal. Predictive probabilities are therefore one possible type of ADM output, not the defining element of the framework.

The practical value of this approach was subsequently illustrated by Godz et al. [1], who instantiated CADEMAs in employee-attrition prevention. Their study did not propose a new CADEMAs variant; rather, it showed how the general architecture can be specialized to a machine-learning setting in which predictive models developed from different organizational perspectives are coordinated with context-dependent management policies to generate prioritizations aligned with organizational objectives.

Although CADEMAs provides a solid conceptual foundation, its adoption is currently limited by the lack of a dedicated software implementation. Researchers and practitioners who wish to apply the framework must still develop their own solutions for routing case data, collecting partial ADM outputs, integrating those outputs, and evaluating the resulting proposal under explicit contextual assumptions. Existing cooperative AI frameworks, multi-agent systems, fuzzy decision-support tools, and hybrid fuzzy-ML applications each address part of this problem, but they do not provide a reusable workflow organized around the CADEMAs separation between decision-unit outputs, integration,

context modulation, and final prioritization. As discussed in Section 2.2, this creates a gap between the conceptual framework and its practical application.

This paper addresses that gap by presenting **CADEMAS-ML**, a software tool that operationalizes the CADEMAs framework for cooperative and context-aware ADM. The main contributions of this work are the following:

- The development of CADEMAs-ML, an open-source software tool that operationalizes the CADEMAs framework for context-aware cooperative automated decision-making using machine-learning models as ADM units.
- The integration of predictive analytics, fuzzy contextual reasoning, and explainable decision support within a unified prioritization process, providing transparent justifications and actionable recommendations for individual cases.
- A robustness-analysis capability that allows decision makers to assess the sensitivity and stability of prioritization outcomes under alternative contextual policies and decision preferences.
- An empirical validation in an employee-attribution prevention scenario involving multiple predictive models and organizational contexts.

The *Explain* and *Robustness* modules are central to this operational contribution. They were not part of the original CADEMAs framework, whose emphasis was architectural, nor were they implemented in the static attrition instantiation of Godz et al. [1]. CADEMAs-ML adds these capabilities to move from a final score to an interactive decision-support workflow in which users can inspect why a case was prioritized and how stable that prioritization is under alternative policy assumptions.

The remainder of the paper is organized as follows. Section 2 reviews the CADEMAs framework, positions it with respect to adjacent cooperative-decision frameworks, and compares application-oriented decision-support systems related to CADEMAs-ML. Section 3 presents CADEMAs-ML, first describing its architecture and methodological improvements and then illustrating its main functionalities through an employee-attribution case study. Finally, Section 4 concludes and outlines directions for future work.

2. Background and related works

2.1. CADEMAs framework

The CADEMAs (Cooperative Automated Decision-Making Systems) framework was introduced as a general structure for designing systems in which multiple automated decision-making units cooperate to produce contextually aligned decisions [7]. The central motivation behind CADEMAs is not limited to prediction. Rather, it is that real-world decision problems, especially in organizational environments, rarely emerge from a single autonomous unit with a single representation of the problem. Decisions may depend on heterogeneous signals produced by classifiers, optimization models, simulation components, rule-based systems, large-language-model agents, or institutional procedures, each with its own assumptions, data access, output type, and reliability level.

To formalize this idea for the prioritization setting addressed in this paper, let $\mathcal{X} = \{x_1, \dots, x_n\}$ denote the case dataset, and let $\mathcal{A} = \{A_1, \dots, A_K\}$ represent a collection of ADM units. In the general CADEMAs framework, each A_k may produce an output $o_k(x_i)$ whose semantics are not necessarily probabilistic: it may be a class, a score, a ranking, a symbolic recommendation, a rule activation, or another decision object. A decision-integration layer is therefore responsible for transforming these heterogeneous outputs into a shared representation and synthesizing them into an integrated proposal. In the ML specialization implemented by CADEMAs-ML, the ADM units are predictive models and the harmonized output is a scalar risk or priority signal $p_k(x_i) \in [0, 1]$. Since these components may differ in performance and trustworthiness, each A_k is assigned a

confidence weight $w_k \geq 0$, with $\sum_{k=1}^K w_k = 1$. These weights can be derived from validation metrics, expert judgment, or governance constraints. The ML specialization aggregates individual outputs into an integrated risk score:

$$R(x_i) = \sum_{k=1}^K w_k p_k(x_i). \quad (1)$$

Rather than treating the integrated output as the final decision criterion, CADEMAs introduces an explicit contextual layer that captures organizational priorities, policies, constraints, and scenario-dependent considerations. The original framework does not prescribe a single context formalism. Context may be represented through crisp rules, utility functions, regulatory constraints, fuzzy propositions, or other evaluative mechanisms. In CADEMAs-ML, and following the attrition application of Godz et al. [1], fuzzy logic is used because it supports gradual degrees of satisfaction instead of binary rules. Let $\mathcal{C} = \{C_1, \dots, C_L\}$ be a set of contextual factors. For each case x_i , the degree to which it satisfies factor C_ℓ is given by $\mu_\ell(x_i) \in [0, 1]$, and these values are aggregated into a contextual score:

$$C(x_i) = H(\mu_1(x_i), \dots, \mu_L(x_i)), \quad (2)$$

where $H : [0, 1]^L \rightarrow [0, 1]$ is an aggregation operator that can take different forms depending on the decision policy (e.g., t-norms, averages, or hierarchical compositions).

The final prioritization index in CADEMAs-ML is obtained by combining the ML-derived risk and the contextual suitability score:

$$\pi(x_i) = \Psi(R(x_i), C(x_i)), \quad (3)$$

where Ψ is a monotonic function that defines the trade-off between both components. A key design principle inherited from CADEMAs is the separation between the integrated ADM proposal and its contextual interpretation. In the ML specialization, this makes explicit what the predictive models suggest and what the organization considers relevant.

The framework has been instantiated in employee attrition prevention by Godz et al. [1]. In that application, multiple machine-learning models were trained on different organizational perspectives (e.g., Human Resources, Finance, Operations, and Employee Wellbeing). Each model acts as an ADM unit within $\mathcal{A}$, and their outputs are combined using weights derived from validation performance, specifically AUC. In parallel, the contextual layer encodes scenario-dependent organizational priorities (e.g., economic crisis or digital transformation settings) using fuzzy membership functions and rule-based structures. This work should therefore be read as an application of CADEMAs to an ML-driven decision-support problem, not as a reformulation of the general CADEMAs framework.

Although this implementation demonstrates the practicality of the framework, it represents only one possible configuration. In general, CADEMAs is designed to remain flexible across several dimensions: the type of ADM units, the semantics of their outputs, the mechanism used for semantic harmonization, the integration operator, and the context-modulation strategy. CADEMAs-ML fixes some of those degrees of freedom in order to provide a concrete software tool: ADM units are H2O [8] MOJO predictive models, the integrated output is a weighted probability-based risk score, the contextual model is expressed through fuzzy rules, and the final decision is a tunable prioritization index. Within that specialization, confidence weights w_k can still be defined using alternative performance measures (e.g., F1-score, precision, recall, calibration metrics, or utility-based indicators), the aggregation operator H may reflect different contextual policies, and the combination function Ψ can be tuned to represent different trade-offs between risk and context.

This flexibility is essential for adapting CADEMAs to heterogeneous organizational environments and motivates the need for a software implementation that operationalizes these design choices in a systematic and configurable way.

2.2. Related frameworks and decision-support applications

Recent research has increasingly recognized that effective decision support requires more than accurate isolated models. In many application domains, decisions emerge from the interaction between automated units, expert knowledge, and contextual considerations that may change over time. For clarity, it is useful to distinguish two related but different bodies of work: general cooperative-decision frameworks, which address how multiple decision entities interact, and application-oriented decision-support systems, which implement concrete methods for a domain or family of decision problems.

At the framework level, Tariq et al. [3] propose an AI-to-collaborator (A2C) framework in which humans and AI systems share decision authority through explicit interaction mechanisms. Similarly, Varga et al. [4] discuss cooperative AI architectures for high-stakes environments and argue that effective collaboration requires a shared representation of context between automated models and domain experts. Other recent systems show related structural patterns in more specific communities: multi-agent LLM planning [9], cooperative driving with LLM-based modules [10], trust-aware multi-agent reinforcement learning [11], distributed decision coordination [12], and ethical multi-agent governance [13]. These works are important alternatives or neighboring frameworks because they address cooperation among decision entities. However, their main focus is not the reusable implementation of a CADEMAs-like workflow in which heterogeneous ADM outputs are explicitly separated from context modulation and final prioritization.

At the application and decision-support level, substantial progress has been made in fuzzy decision support, multi-criteria decision making, and hybrid fuzzy-ML workflows. Applications such as MAKEDECISION [14] and the fuzzy DEMATEL-based DSS described by Chekry et al. [15] emphasize transparency and traceability by exposing intermediate calculations to users. A related stream has explored the combination of machine learning and fuzzy reasoning. The survey by Kumar et al. [5] highlights the growing interest in hybrid fuzzy-ML systems and identifies configurability and expert participation as key challenges for future decision-support tools. Other studies have proposed application-specific integrations of predictive models and expert knowledge, including fuzzy decision-support architectures [6], Fermatean fuzzy decision frameworks [16], and context-sensitive prioritization approaches based on fuzzy AHP and machine learning [17]. These contributions demonstrate the value of combining data-driven evidence with contextual or expert-defined criteria, but most are tailored to particular methods or domains rather than organized around the CADEMAs decomposition of ADM units, integration, context, and final prioritization.

Open-source fuzzy and MCDM libraries, including PYFDM [18], PYIFDM [19], FuzzyR [20], the intuitionistic-fuzzy toolkit of Roszkowska and Wachowicz [21], the ELIGERE platform [22], and the Fuzzy Cognitive Maps package presented by Dikopoulou and Papageorgiou [23], are also relevant. They improve reproducibility and provide reusable computational building blocks. Nevertheless, software availability is not the central distinction pursued here. The key issue is whether a system captures the cooperative ADM problem that CADEMAs is intended to solve: multiple decision-producing units, explicit integration of their outputs, contextual evaluation that is kept conceptually separate, and a final decision mechanism whose assumptions can be inspected and varied.

Table 1 summarizes representative application-oriented contributions in these areas. Two observations emerge from this comparison. First, several works provide valuable

Table 1. Comparison of representative application-oriented decision-support works related to CADEMAs-ML. The comparison emphasizes whether each contribution captures the cooperative ADM elements targeted by CADEMAs, rather than whether software is publicly released.

Work	Decision-support focus	Multiple ADM units	Explicit context layer	Relation to CADEMAs-ML
Godz et al. [1]	Employee-attribution prevention using CADEMAs with departmental ML models	Yes	Yes	Direct application that motivates the implemented workflow
Więckowski and Sałabun [14]	Auditable fuzzy MCDM through a web-based tool	No	Partly	Strong transparency, but no cooperative ADM integration layer
Chekry et al. [15]	Fuzzy DEMATEL decision-support system	No	Partly	Traceable fuzzy causal analysis, but method-specific rather than CADEMAs-like
Ghosh et al. [6]	Fuzzy DSS with integrated expert rules for predictive workflows	Partly	Yes	Combines prediction and expert rules, but as a bespoke workflow
Aly et al. [17]	Context-sensitive prioritization using fuzzy AHP and ML	Partly	Yes	Hybrid fuzzy-ML prioritization without a general ADM architecture
Suvitha et al. [16]	Fermatean fuzzy aggregation for expert-driven preferences	No	Partly	Rich aggregation operators, but no predictive/ADM cooperation layer
CADEMAs-ML (this work)	Configurable prioritization with ML-based ADM units and fuzzy context scenarios	Yes	Yes	Operationalizes CADEMAs for reusable ML-driven decision support

fuzzy, MCDM, or hybrid fuzzy-ML mechanisms for transparent decision support. Second, these mechanisms usually address one part of the CADEMAs problem: preference aggregation, explainable fuzzy rules, contextual scoring, or predictive assistance. What remains missing is an operational workflow in which multiple ADM units are treated as cooperating components, their outputs are integrated explicitly, contextual evaluation is modeled as a separate layer, and the final prioritization can be recomputed under different decision policies.

CADEMAs-ML is designed to address this gap for the predictive-model setting. Building upon the conceptual framework introduced by Novoa-Hernández et al. [7] and the employee-attribution instantiation reported by Godz et al. [1], it provides a configurable software implementation in which ML-based ADM outputs and contextual reasoning coexist as explicit and auditable components of the decision-making process. Its planned open-source availability is an additional benefit, but the main contribution is the operationalization of the CADEMAs decomposition in a reusable decision-support tool.

3. CADEMAs-ML software tool

This section presents CADEMAs-ML as a software realization of one concrete CADEMAs specialization: cooperative prioritization supported by multiple predictive ADM units and an explicit fuzzy context layer. We first explain how the general CADEMAs

architecture is instantiated in a configurable ML workflow. Special attention is given to the fuzzy context model, the case-level *Explain* module, and the robustness analysis of the prioritization ranking. We then use the employee-attrition example distributed with the repository to show the main functionalities of the application.

3.1. Architecture and improvements

CADEMAS-ML is not a domain-specific attrition script. It is an operational implementation of CADEMAs for the case in which the cooperating ADM units are predictive models. In line with Novoa-Hernández et al. [7], the software keeps partial automated assessments separate from their contextual interpretation. Its contribution is to make this separation executable for ML-based decision support. Predictive components, contextual rules, and decision policies are configured explicitly and can be inspected through the interface. Thus, the attrition study of Godz et al. [1] becomes one possible configuration of the tool, not its only intended use.

The application is implemented in Python and combines a Streamlit [24] front end, H2O MOJO inference, and a vectorized fuzzy-context engine. Interactive visual analytics are rendered with Altair [25] and embedded in the *Overview*, *Context*, *Explain*, and *Robustness* modules. Because Altair produces declarative chart specifications, the resulting figures—including scatter plots, membership-function views, attribution bar charts, and the three robustness summaries introduced below—can be exported directly for reporting and publication. A run starts from four inputs: the case dataset, the MOJO models, a model-metadata JSON file, and one or more context JSON files. These inputs map naturally onto the CADEMAs layers. The dataset and configuration files act as the input manager. The MOJO files instantiate the ADM units in this ML specialization. Their weighted probabilistic outputs form the integrated predictive risk. The fuzzy-rule engine provides the contextual model. The final layer combines both streams through a user-adjustable policy parameter. Figure 1 summarizes this architecture.

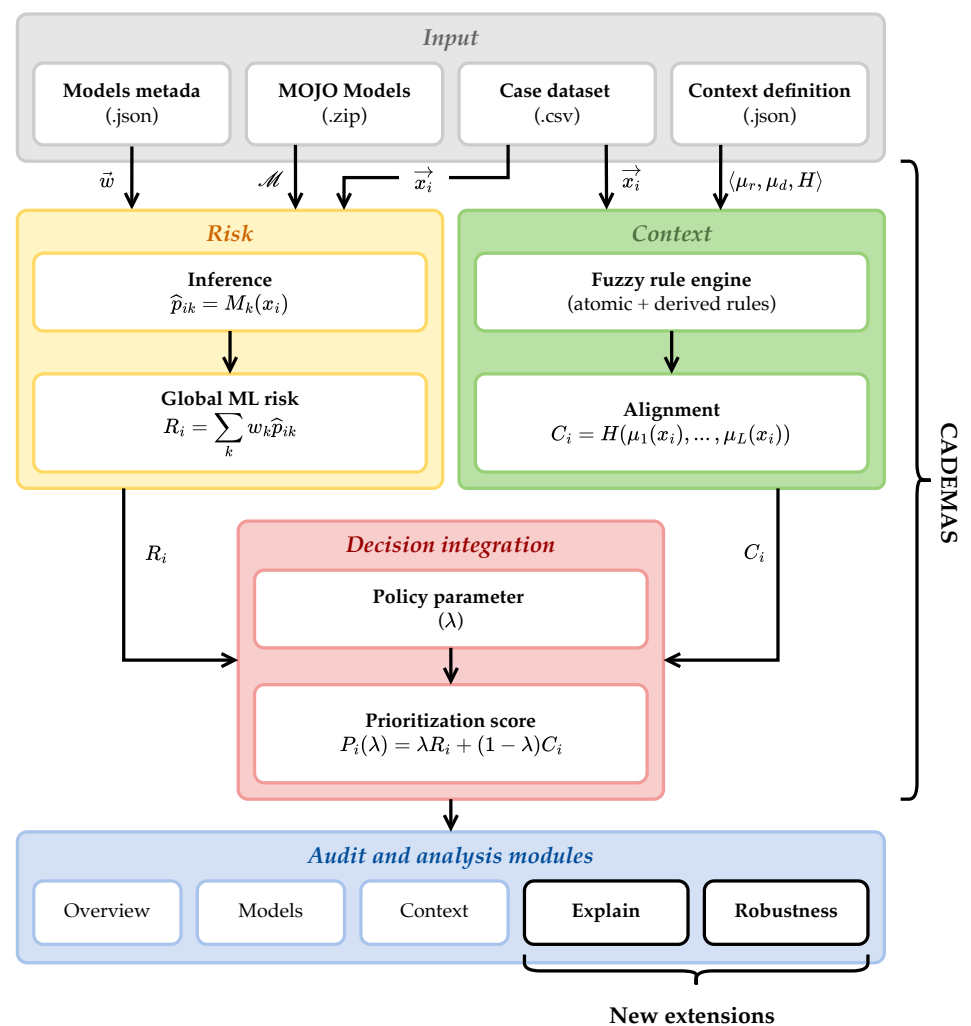


Figure 1. Operational architecture of CADEMAs-ML application. The software instantiates the CADEMAs [7] architecture as a configurable vertical pipeline in which predictive risk and contextual alignment are computed independently before being integrated into a prioritization index.

3.1.1. ML-based cooperative risk

The predictive layer follows the cooperative aggregation principle of CADEMAs as specialized in the attrition instantiation of Godz et al. [1]. CADEMAs-ML exposes that specialization as a reusable software component. Let $\mathcal{X} = \{x_1, \dots, x_n\}$ be the case dataset and let $\mathcal{M} = \{M_1, \dots, M_K\}$ be the uploaded predictive ADM units. Each model M_k produces, for case x_i , a positive-class probability

$$\hat{p}_{ik} \in [0, 1].$$

The models need not use the same feature subset. Each MOJO receives the available case matrix and uses its internal schema to select the required variables. The contribution of each component is controlled by a non-negative confidence weight w_k . This weight is computed from a user-selected validation metric s_k :

$$w_k = \frac{s_k}{\sum_{\ell=1}^K s_\ell}, \quad \sum_{k=1}^K w_k = 1. \quad (4)$$

The cooperative machine-learning risk score is then

$$R_i = \sum_{k=1}^K w_k \hat{p}_{ik}. \quad (5)$$

This mathematical form is the ML-specific realization of the more general CADEMAs decision-integration layer. The improvement introduced by CADEMAs-ML is practical configurability. Users can upload any number of compatible MOJO models, read model metadata from JSON, and choose the validation metric used for weighting.

3.1.2. Configurable fuzzy context model

The context layer is where CADEMAs-ML most clearly turns the conceptual framework into an operational artifact. The role of context follows Novoa-Hernández et al. [7], and the use of fuzzy propositions follows the attrition application in Godz et al. [1]. The difference is that CADEMAs-ML does not hard-code those contexts. It represents them as JSON specifications. The same formalism can therefore be reused with other domains, rule families, and decision policies.

Formally, a context is defined as a collection of fuzzy propositions. Let V_r be a case attribute and let $\tilde{F}_r$ be a fuzzy set representing a linguistic predicate over the universe of V_r . For case x_i , an atomic rule r evaluates the proposition “ V_r is $\tilde{F}_r$ ” as

$$\mu_r(x_i) = \mu_{\tilde{F}_r}(x_{ir}) \in [0, 1], \quad (6)$$

where x_{ir} denotes the value of attribute V_r for case x_i . The implementation supports triangular, trapezoidal, linear increasing, and linear decreasing membership functions. It also supports categorical maps and categorical sets. The former assign degrees directly to symbolic values. The latter group categories into linguistic labels that can be scored later in derived rules.

Atomic rules may include activation conditions. If $m_r(x_i) \in \{0, 1\}$ denotes whether the when condition of rule r is satisfied for case x_i , the effective membership is

$$\mu_r^*(x_i) = m_r(x_i) \mu_r(x_i), \quad (7)$$

with $m_r(x_i) = 1$ for unconditional rules. This makes it possible to represent policies such as department-specific salary suitability without duplicating datasets or writing procedural code.

Beyond atomic rules, CADEMAs-ML allows the definition of derived rules. A derived rule d combines a subset of already computed memberships through an aggregation operator G_d :

$$\mu_d(x_i) = G_d(\mu_{r_1}^*(x_i), \dots, \mu_{r_q}^*(x_i)). \quad (8)$$

The implemented operators are minimum-based AND, maximum-based OR, algebraic PRODUCT, and arithmetic AVERAGE. The context alignment score is then obtained by applying a final aggregation policy H to selected atomic and derived memberships:

$$C_i = H(\mu_1(x_i), \dots, \mu_L(x_i)). \quad (9)$$

The function H can be selected interactively or specified as a declarative logic tree in the context JSON. This captures the attrition instantiation in Godz et al. [1], but makes it portable. It also lets organizations express conservative, compensatory, hierarchical, or conditional contextual policies without changing the application code. The complete rule

and membership-function specifications for the context addressed in the case study are reported in Appendix B.

All atomic and derived memberships are retained as an audit matrix. The final context score is therefore not a black-box modifier of the ML risk. It can be decomposed into the rule activations that produced it.

3.1.3. Decision integration and robustness

Predictive evidence and contextual suitability remain separate until the final decision stage. For a fixed policy parameter $\lambda \in [0, 1]$, CADEMAs-ML computes the prioritization score as

$$P_i(\lambda) = \lambda R_i + (1 - \lambda)C_i. \quad (10)$$

The interpretation is direct. A value $\lambda = 1$ produces a risk-driven ranking. A value $\lambda = 0$ produces a context-driven ranking. Intermediate values encode explicit trade-offs between both components.

The robustness module is a specific improvement of CADEMAs-ML over the previous static attrition instantiation. Let $\Lambda = \{\lambda_0, \dots, \lambda_q\}$ be a finite grid of policy values. For each $\lambda \in \Lambda$, the tool computes the score vector $P(\lambda)$ and the ranking

$$r_i(\lambda) = 1 + |\{j : P_j(\lambda) > P_i(\lambda)\}|, \quad (11)$$

with deterministic tie-breaking by case order. This produces a rank trajectory for every case. CADEMAs-ML summarizes these trajectories through three complementary Altair views: a bump chart, a ranks box plot, and a rank-acceptability heatmap. The bump chart shows how each case moves as λ changes. The box plot summarizes the distribution of ranks over Λ ; cases are ordered by median rank and interquartile range. Finally, the rank acceptability index estimates how often a case attains a given rank:

$$b_i^r = \frac{1}{|\Lambda|} \sum_{\lambda \in \Lambda} \mathbf{1}\{r_i(\lambda) = r\}. \quad (12)$$

Together, these indicators distinguish stable priorities from policy-sensitive cases.

3.1.4. Case-level explanation and natural-language recommendations

A second improvement introduced by CADEMAs-ML is the *Explain* module. The original CADEMAs framework specifies the separation between ADM outputs, context evaluation, and final decision integration, but it does not prescribe a case-level explanation mechanism. Similarly, the attrition instantiation of Godz et al. [1] reports contextual prioritization outcomes, but does not provide an interactive module that explains individual cases in natural language. CADEMAs-ML fills this gap by storing intermediate artifacts during the analysis run and exposing them through a four-part local explanation.

For a selected case x_i , the first explanation level decomposes the final score in Equation (10) into its two additive components:

$$P_i(\lambda) = \underbrace{\lambda R_i}_{\text{predictive contribution}} + \underbrace{(1 - \lambda)C_i}_{\text{contextual contribution}}. \quad (13)$$

This decomposition makes explicit whether the prioritization is mainly driven by predictive risk, contextual alignment, or a balance between both.

The second level provides local ML attribution. Since the uploaded H2O AutoML MOJO models are treated as black-box predictive ADM units and may include stacked ensembles, CADEMAs-ML uses a model-agnostic perturbation procedure rather than

relying on model-specific SHAP values. For each model M_k and feature f , the tool compares the original predicted probability with the prediction obtained after replacing the value of f by a cohort baseline b_f (median for numerical variables and mode for categorical variables):

$$\Delta_{ikf} = \hat{p}_{ik}(x_i) - \hat{p}_{ik}(x_i^{(f \leftarrow b_f)}). \quad (14)$$

The model-specific effects are aggregated with the same confidence weights used for the cooperative risk:

$$\Delta_{if} = \sum_{k=1}^K w_k \Delta_{ikf}. \quad (15)$$

Positive values indicate features whose observed value increases the local predicted risk relative to the cohort baseline, while negative values indicate features whose observed value decreases it. These quantities are expressed in probability points and should be interpreted as approximate local effects, not as causal impacts.

The third level traces the contextual component by presenting the fuzzy memberships $\mu_r(x_i)$ of the atomic and derived rules that produced C_i . This connects the numerical context score with the policy rules encoded in the JSON context specification. Finally, the fourth level generates a deterministic natural-language summary. The text combines the priority score, the balance parameter λ , the risk and context levels, the most relevant ML drivers, the most relevant contextual drivers or bottlenecks, and a generic tier-based recommendation. The recommendation uses the final score P_i : *Tier 1* for high priority, *Tier 2* for medium or borderline priority requiring manual qualitative review, and *Tier 3* for low priority under the current prediction and context guidelines.

3.2. Main functionalities

We illustrate the main functionalities of CADEMAs-ML through an employee-attribution scenario that reproduces the methodological setup of Godz et al. [1] as an interactive software workflow. The objective is not to improve attrition prediction accuracy, but to show how cooperative predictive assessments and explicit contextual policies are configured, executed, audited, and explained within the architecture described in Section 3.1.

The scenario assumes that an organization must prioritize retention interventions among employees at attrition risk when resources are limited. Predictive evidence is not monolithic: we simulate four organizational perspectives—Culture and Wellbeing, Finance, Operations and Development, and Human Resources—each modeled as an independent ADM unit that observes only a subset of employee attributes from the same workforce records. This hypothetically reflects departments with partial, perspective-specific access to human-resources data. All units were trained offline on the publicly available Kaggle dataset *IBM HR Analytics Employee Attrition and Performance*¹, which contains 1,470 employee records and 35 attributes describing demographic, professional, compensation, performance, and organizational conditions. Following Godz et al. [1], H2O AutoML was run independently for each feature subset defined in Table A1, using an 80/20 train-validation partition and a one-hour time budget per model. The selected MOJO artifacts, their validation metrics, and feature assignments are distributed with the application. At run time, CADEMAs-ML loads these models, produces case-level probabilities, and aggregates them into the cooperative risk score R_i using the AUC-based weights in Equation (4).

Contextual prioritization is encoded separately through JSON fuzzy specifications. The example bundle includes two alternative organizational policies: *Economic Crisis* and

¹ <https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset> (accessed 19 March 2026).

Figure placeholder: *Home* module and sidebar upload controls for model definitions, context configurations, MOJO models, and the case dataset.

Figure 2. *Home* and input configuration module. The user uploads or loads the four inputs and launches the analysis pipeline described in Section 3.1.

Digital / Technological Transformation. They share the same predictive layer but differ in the rules used to compute C_i . In the walkthrough below, we focus on *Digital / Technological Transformation*, which prioritizes upskilling capacity, role adaptability, career scalability, and investment horizon during a technological-change programme. The complete rule set for this context is reported in Appendix B.

For interactive demonstration, the repository provides a 20-case CSV subset with the same 35 columns as the original dataset, including the attrition outcome label. This label is retained for verification but is not used as a prioritization criterion. Employee names in *Case_ID* were randomly generated to simplify reference in the interface and in this paper. The subsections that follow describe how CADEMAs-ML processes these inputs through its interface modules: loading configuration, computing R_i and C_i , integrating them into $P_i(\lambda)$, and supporting inspection through the *Overview*, *Models*, *Context*, *Explain*, and *Robustness* views.

3.2.1. Input configuration and analysis workflow

The entry point is the *Home* module and the sidebar upload controls (Figure 2). The user provides the four inputs required by the architecture—model-metadata JSON, context JSON, MOJO models, and the case dataset—or loads the ready-made example distributed under `example_attrition`². The application *Help* panel documents the same bundle. Before execution, the user selects the validation metric used for model weighting, chooses one of the uploaded context specifications, sets the balance parameter λ , and triggers *Run analysis*. CADEMAs-ML then executes the pipeline defined in Section 3.1: MOJO inference, fuzzy context evaluation, and score integration. Table 2 summarizes the four pre-trained ADM units included in the example and the AUC-based weights recomputed during the run.

Following the training protocol of Godz et al. [1], the four MOJO models correspond to the organizational perspectives summarized in Table 2. Using AUC as the weighting metric, the reproduced run assigns a lower weight to Operations and Development because its validation AUC is lower than that of the other three components.

3.2.2. Overview: prioritized cases

After execution, the *Overview* module surfaces the outputs of Equations (5)–(10). It presents global indicators, a risk–context scatter plot, the score distribution, and the

² https://github.com/pnvoa/cademas-app/tree/main/example_attrition.

Table 2. Models used in the attrition example and AUC-based weights computed by CADEMAs-ML. The *AutoML model* column summarizes the artifact exported by H2O AutoML.

ID	Organizational perspective	AutoML model	AUC	Weight
M_1	Culture and Wellbeing	Stacked Ensemble (BestOfFamily)	0.9941	0.2623
M_2	Finance	Stacked Ensemble (BestOfFamily)	0.9972	0.2631
M_3	Operations and Development	GBM (grid 1, model 26)	0.8022	0.2116
M_4	Human Resources	Stacked Ensemble (BestOfFamily)	0.9968	0.2630

Figure placeholder: *Overview* module showing global metrics, the risk-context scatter plot, prioritization-score distribution, and prioritized cases table.

Figure 3. *Overview* module for the attrition example. The module summarizes global indicators, the risk-context scatter plot, the score distribution, and the prioritized cases table ordered by P_i .

prioritized cases table ordered by P_i . In the reproduced *Digital / Technological Transformation* context, C_i ranged from 0.1500 to 0.7100, with a mean of 0.4663. Table 3 reports the ten cases with the highest prioritization score $P_i(0.5)$ under the default balance parameter $\lambda = 0.5$, together with the cooperative risk R_i and context alignment C_i that compose that score. The same ten cases are used later in the robustness analysis of Figure 7.

The ranking in Table 3 is driven primarily by cooperative predictive risk. All ten cases exhibit high R_i , while C_i varies more widely. *Evelyn Taylor* leads the table because both components are high ($R_i = 0.9555$, $C_i = 0.7000$). Cases such as *Lucas Wright* and *Abigail Anderson* remain among the top ten mainly because of critical predictive risk, even though their contextual alignment is moderate or low. By contrast, *Noah Lewis*—the most context-aligned case in the cohort ($C_i = 0.7100$)—does not appear in the table because $R_i = 0.0651$ is low, yielding $P_i(0.5) = 0.3875$. This illustrates why CADEMAs-ML reports P_i , R_i , and C_i separately rather than ranking cases by contextual alignment alone.

3.2.3. *Models*: inspecting cooperative risk

The *Models* module exposes the intermediate results of the cooperative risk layer. It displays the normalized model weights, the per-case probabilities returned by each MOJO, and the aggregated score R_i .

3.2.4. *Context*: auditing fuzzy policies

The *Context* module renders the fuzzy audit underlying Equation (9). It shows membership functions, atomic and derived rule activations, and the resulting C_i values for each case. In the reproduced *Digital / Technological Transformation* run, C_i ranged from 0.1500 to 0.7100.

Table 3. Top prioritized employees in the *Digital / Technological Transformation* context at $\lambda = 0.5$, ranked by $P_i(0.5)$.

Rank	Case	$P_i(0.5)$	R_i	C_i
1	<i>Evelyn Taylor</i>	0.8277	0.9555	0.7000
2	<i>Amelia Davis</i>	0.7897	0.9879	0.5914
3	<i>Theodore Adams</i>	0.7101	0.9803	0.4400
4	<i>William Young</i>	0.7101	0.9751	0.4450
5	<i>Benjamin Scott</i>	0.7086	0.9622	0.4550
6	<i>Charlotte Miller</i>	0.6727	0.9204	0.4250
7	<i>Lucas Wright</i>	0.6616	0.9882	0.3350
8	<i>Harper Wilson</i>	0.6170	0.9641	0.2700
9	<i>Henry King</i>	0.5688	0.9676	0.1700
10	<i>Abigail Anderson</i>	0.5648	0.9796	0.1500

Figure placeholder: *Models* module with AUC-based weights and per-case MOJO probabilities contributing to the Global ML Risk score.

Figure 4. *Models* module. The table links each uploaded MOJO to its confidence weight and displays the per-case probability outputs used to compute R_i .

3.2.5. *Explain*: local case interpretation

For a selected case, the *Explain* module presents the four explanation levels defined in Section 3.1: score decomposition, local ML attributions, fuzzy-rule traceability, and a deterministic natural-language summary (Figure 6).

Two cases from the *Digital / Technological Transformation* run illustrate the interpretation produced by the *Explain* module at $\lambda = 0.5$. The first is *Evelyn Taylor*, ranked first in Table 3; the second is *Noah Lewis*, who is excluded from that table despite having the highest C_i in the cohort. The summaries below were reproduced from CADEMAs-ML and follow its deterministic template: the final prioritization score P_i is reported first, then λ , the levels of R_i and C_i , the main ML drivers, the contextual rules that support or constrain priority, and a tier-based recommendation derived from P_i .

Evelyn Taylor combines critical predictive risk ($R_i = 0.9555$) with high contextual alignment ($C_i = 0.7000$), yielding $P_i(0.5) = 0.8277$ and a *Tier 1* recommendation:

The case prioritization score for *Evelyn Taylor* is 83%. This decision reflects a balanced posture ($\lambda = 0.50$), combining critical predictive risk (96%) with high contextual alignment (70%). On the predictive side, risk was driven mainly by Department (Sales) and Overtime (Yes). In context, the case aligns strongly with the policy rules, particularly on Relevant years at company and Age investment window.

Suggestion [*Tier 1* – High Priority]: Recommended to proceed with intervention or resource allocation immediately.

Noah Lewis provides a contrasting interpretation. Although this case has the highest contextual alignment in the cohort ($C_i = 0.7100$), its cooperative risk is low ($R_i = 0.0651$).

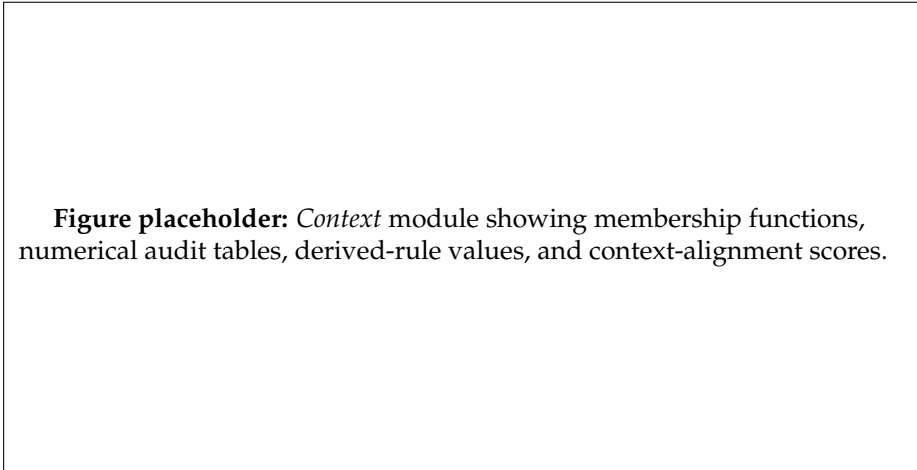


Figure 5. *Context* module. Atomic and derived membership values are exposed for audit, allowing analysts to inspect why a case receives a given contextual score.

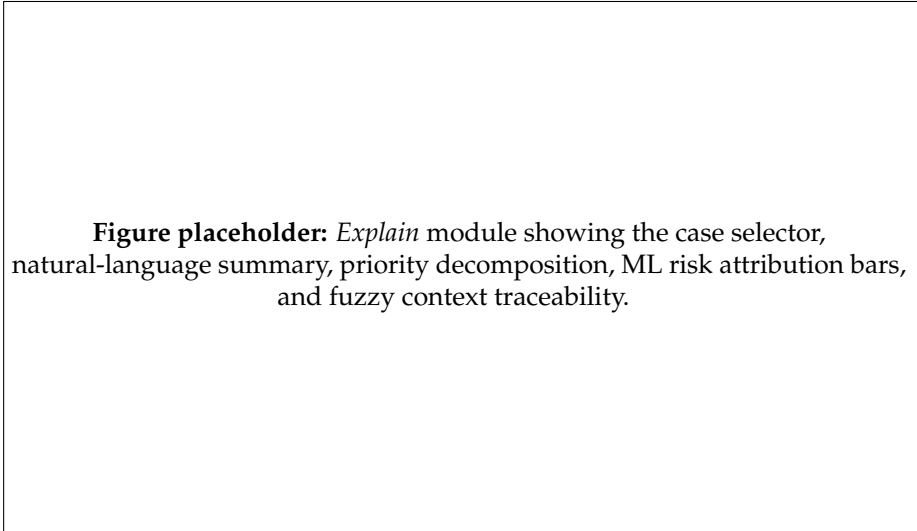


Figure 6. *Explain* module. CADEMAs-ML provides local explanations for a selected case by combining score decomposition, approximate ML attributions, fuzzy-rule traceability, and a natural-language recommendation.

The resulting prioritization score is therefore $P_i(0.5) = 0.3875$, and the module assigns *Tier* 3:

The case prioritization score for *Noah Lewis* is 39%. This decision reflects a balanced posture ($\lambda = 0.50$), combining low predictive risk (7%) with high contextual alignment (71%). On the predictive side, risk was driven mainly by Years with current manager (8) and Number of companies worked (1). In context, the case aligns strongly with the policy rules, particularly on Relevant years at company and Age investment window. Suggestion [*Tier 3 – Low Priority*]: Intervention or resource allocation is not justified under current prediction and context guidelines.

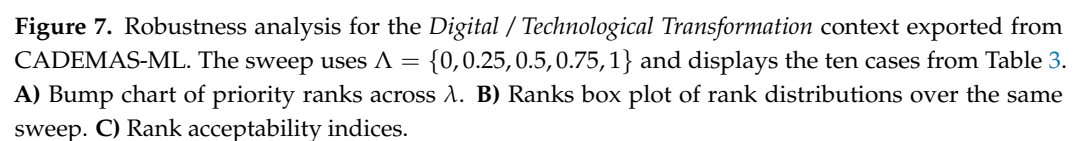
In practical terms, CADEMAs-ML explains not only why a case is prioritized, but also why another case with strong contextual alignment may still receive a low final score when predictive risk remains limited. A third illustrative pattern is provided by *Theodore Adams* ($R_i = 0.9803$, $C_i = 0.4400$, $P_i(0.5) = 0.7101$), who receives *Tier 2* because high predictive risk and only moderate contextual alignment produce borderline evidence under the current policy.

3.2.6. Robustness: sensitivity of the priority ranking

The *Robustness* module implements the sensitivity analysis defined in Section 3.1. Figure 7 illustrates the exported Altair charts for the *Digital / Technological Transformation* run using $\Lambda = \{0, 0.25, 0.5, 0.75, 1\}$ and the same ten cases reported in Table 3.

Fig. 7-A) shows the bump chart of priority ranks across λ . Fig. 7-B) reports the ranks box plot, which summarizes the rank distribution of each displayed case over the same sweep; cases are ordered from left to right by median rank and, in case of ties, by interquartile range. Fig. 7-C) shows the rank acceptability indices, estimating the relative frequency with which each case occupies each rank position over Λ .

When λ approaches 0 and contextual alignment receives greater weight, cases outside Table 3 such as *Noah Lewis* and *Emma Smith* rise in Fig. 7-A) because they combine high C_i with low R_i . When λ approaches 1, the high-risk cases already listed in the table—such as *Lucas Wright* and *Theodore Adams*—dominate the ranking regardless of transformation alignment. *Evelyn Taylor* remains near the top across intermediate values of λ because both R_i and C_i are high. The box plot in Fig. 7-B) makes this contrast visible at a glance: stable high-priority cases exhibit compact rank distributions, whereas policy-sensitive cases show wider spreads. Fig. 7-C) complements both views by quantifying how often each case attains each rank over the λ grid.



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ommendations. These capabilities were not present in the original CADEMAs framework and were not implemented in the attrition instantiation of Godz et al. [1].

The employee-attrition case study illustrates how the same set of predictive models can lead to different prioritization outcomes depending on the contextual policy adopted. It also shows how individual cases can be inspected before action is taken. Future work will focus on broadening the supported ADM-unit types beyond predictive MOJO models, incorporating additional explanation methods when model internals are available, improving governance support for context-versioning and audit trails, and validating the tool in other prioritization domains such as fraud triage, credit review, resource allocation, and medical or social-service prioritization.

References

- Godz, M.; Novoa-Hernández, P.; Pelta, D.A., A Cooperative and Context-Aware Approach for Employee Attrition Prevention. In *Information Processing and Management of Uncertainty in Knowledge-Based Systems*; Springer Nature Switzerland, 2026; pp. 203–217. https://doi.org/10.1007/978-3-032-29000-7_15.
- ELI Innovation Paper. Guiding Principles for Automated Decision-Making in the EU. Innovation paper, 2022.
- Tariq, M.; et al. From AI to Collaborator: A Framework for Human-AI Cooperative Decision-Making. *Information Fusion* **2024**. Conceptual framework, no public software.
- Varga, P.; et al. Cooperative AI Architectures for High-Stakes Decision-Making: A Review. *Artificial Intelligence Review* **2023**.
- Kumar, A.; et al. Hybrid Fuzzy-Machine Learning Architectures for Uncertainty-Aware Decision Support: A Survey. *Information Sciences* **2025**.
- Ghosh, S.; et al. A Fuzzy Decision Support System with Integrated Expert Rules for Predictive Workflows. *Applied Soft Computing* **2020**.
- Novoa-Hernández, P.; Pelta, D.A.; Godz, M.; Verdegay, J.L.; Buendia-Carrillo, D. CADEMAs – A Framework for Cooperative Automated Decision-Making Systems. In Proceedings of the 2026 IEEE Conference on Artificial Intelligence (CAI), 2026, pp. 756–761. <https://doi.org/10.1109/CAI68641.2026.11536392>.
- H2O.ai. H2O.ai: The Open Source AI Platform, 2026. Accessed: 2026-01-12.
- Cai, X.; Meng, Z.; Guo, Q.; Jiang, L.; Wu, J.; Min, M.; Cheng, Y.; Gui, X.; Zhao, J. Mingyue: A Multi-Agent System for Human-AI Collaborative Decision-Making Under Safety Constraints in Complex Power Grids. In Proceedings of the 2025 5th Power System and Green Energy Conference (PSGEC), 2025, pp. 434–439. <https://doi.org/10.1109/PSGEC66102.2025.11150990>.
- Fang, S.; Liu, J.; Ding, M.; Cui, Y.; Lv, C.; Hang, P.; Sun, J. Towards Interactive and Learnable Cooperative Driving Automation: A Large Language Model-Driven Decision-Making Framework. *IEEE Transactions on Vehicular Technology* **2025**, *74*, 11894–11905. <https://doi.org/10.1109/TVT.2025.3552922>.
- Pan, J.; Wang, T.; Claudel, C.; Shi, J. Trust-MARL: Trust-Based Multi-Agent Reinforcement Learning Framework for Cooperative On-Ramp Merging Control in Heterogeneous Traffic Flow, 2025, [arXiv:cs.MA/2506.12600].
- Ibro, H.; Mahala, G.; Pulawski, S.; Harvey, S.; Miller, A.A.; Ghose, A.; Dam, H.K. DITURIA: A Framework for Decision Coordination Among Multiple Agents. In Proceedings of the PRIMA 2022: Principles and Practice of Multi-Agent Systems: 24th International Conference Proceedings. Springer-Verlag, 2022, p. 492–508. https://doi.org/10.1007/978-3-031-21203-1_29.
- Lakshmana Perumal, R.K. Holistic Ethical Commons: A Dynamic Framework for Ethical Decision Making in Multi-Agent AI Systems. In Proceedings of the 2025 5th Intelligent Cybersecurity Conference (ICSC), 2025, pp. 378–384. <https://doi.org/10.1109/ICSC65596.2025.11140371>.
- Więckowski, J.; Sałabun, W. MakeDecision: A Web-Based Tool for Fuzzy Multi-Criteria Decision Making with Auditable Intermediate Calculations. *SoftwareX* **2024**.
- Chekry, A.; et al. A Fuzzy DEMATEL-Based Decision Support System with Traceable Intermediate Computations. *Expert Systems with Applications* **2024**.

16. Suvitha, K.; et al. Fermatean Fuzzy Decision Operators for Expert-Driven Preference Aggregation. *International Journal of Fuzzy Systems* **2025**. 584
17. Aly, M.; et al. Integrating Fuzzy AHP with Machine Learning for Context-Sensitive Prioritization. *Expert Systems with Applications* **2018**. 585
18. Więckowski, J.; Kizielewicz, B.; Sałabun, W. pyFDM: A Python Library for Uncertainty Decision Analysis Methods. *SoftwareX* **2022**, *20*, 101271. Open-source Python package. 586
19. Więckowski, J.; Sałabun, W. pyIFDM: A Python Module for Intuitionistic Fuzzy Multi-Criteria Decision Making. *SoftwareX* **2023**. Open-source Python package. 587
20. Chen, C.; Wagner, C.; Garibaldi, J.M. FuzzyR: An R Package for Fuzzy Inference Systems. *Journal of Statistical Software* **2020**. Open-source R package. 588
21. Roszkowska, E.; Wachowicz, T. An Open-Source Intuitionistic-Fuzzy MCDM Toolkit for Group Decision Support. *Group Decision and Negotiation* **2024**. 589
22. Grazioso, S.; et al. Eligere: An Open-Source Platform for Fuzzy Electoral and Ranking Applications. *SoftwareX* **2023**. 590
23. Dikopoulou, Z.; Papageorgiou, E.I. A Python Package for Fuzzy Cognitive Maps Modeling and Analysis. *SoftwareX* **2021**. Open-source Python package. 591
24. Streamlit Inc.. Streamlit. Software, 2026. Version 1.53.0. 592
25. VanderPlas, J.; Granger, B.; Heer, J.; Moritz, D.; Wongsuphasawat, K.; Satyanarayan, A.; Lees, E.; Timofeev, I.; Welsh, B.; Sievert, S. Altair: Interactive Statistical Visualizations for Python. *Journal of Open Source Software* **2018**, *3*, 1057. <https://doi.org/10.21105/joss.01057>. 593

Appendix A. Attrition example dataset and model features

The 20-case attrition example distributed with CADEMAs-ML contains 36 columns: Case_ID, 34 predictor variables, and the attrition outcome label. The predictors cover demographic, professional, compensation, performance, and organizational attributes from the *IBM HR Analytics Employee Attrition and Performance* Kaggle dataset used in Godz et al. [1]. Five variables (age, gender, department, job_role, and job_level) are shared by all four MOJO models. Two additional columns (employee_count and employee_number) are retained in the CSV file but are not used by any model in the example bundle. Each MOJO receives the full case matrix and selects its required variables according to the schema stored in the model-definition JSON file.

Table A1 summarizes the feature assignment. The Human Resources model reads over18 from the CSV file; the corresponding field is named over_18 in the model metadata.

Table A1. Predictor variables in the attrition example and the MOJO models that use each one. M_1 – M_4 denote Culture and Wellbeing, Finance, Operations and Development, and Human Resources, respectively.

Predictor	M_1	M_2	M_3	M_4
age	✓	✓	✓	✓
business_travel	✓			
daily_rate		✓		
department	✓	✓	✓	✓
distance_from_home				✓
education				✓
education_field				✓
employee_count				
employee_number				
environment_satisfaction	✓			
gender	✓	✓	✓	✓
hourly_rate		✓		
job_involvement	✓			
job_level	✓	✓	✓	✓
job_role	✓	✓	✓	✓
job_satisfaction	✓			
marital_status				✓
monthly_income		✓		
monthly_rate		✓		
num_companies_worked			✓	
over18				✓
over_time	✓			
percent_salary_hike		✓		
performance_rating			✓	
relationship_satisfaction	✓			
standard_hours				✓
stock_option_level		✓		
total_working_years			✓	
training_times_last_year			✓	
work_life_balance	✓			
years_at_company			✓	
years_in_current_role			✓	
years_since_last_promotion			✓	
years_with_curr_manager			✓	

Appendix B. Context specifications for the attrition scenarios

The example repository includes two context configurations, but the case study focuses on the *Digital / Technological Transformation* context. This appendix reports the corresponding declarative JSON specification in compact form. The rules are examples of the generic formalism described in Section 3.1; they are not hard-coded assumptions of CADEMAs-ML.

Tables A2 and A3 summarize the atomic rules, derived rules, and final aggregation policy used in the case study.

Table A2. Atomic fuzzy rules in the *Digital / Technological Transformation* context.

Rule	Feature	Type	Parameters
training_high	training_times_last_year	linear increasing	(1,6)
education_high	education	linear increasing	(2,5)
years_company_relevant	years_at_company	triangular	(3,10,20)
years_role_relevant	years_in_current_role	triangular	(1,6,11)
age_investment_window	age	trapezoidal	(22,28,50,57)
experience_sufficient	total_working_years	linear increasing	(2,10)
role_adaptability	job_role	categorical map	Research Scientist = 0.9, Research Director = 0.9, Manager = 0.8, Manufacturing Director = 0.7, Healthcare Representative = 0.7, Sales Executive = 0.6, Laboratory Technician = 0.4, Sales Representative = 0.4, Human Resources = 0.5.

Table A3. Derived rules and final aggregation in the *Digital / Technological Transformation* context.

Rule or policy	Operator	Inputs and role
upskilling_capacity	AVERAGE	training_high and education_high.
career_scalability	PRODUCT	years_company_relevant and years_role_relevant.
investment_horizon	PRODUCT	age_investment_window and experience_sufficient.
Final H	AND with average aggregation	upskilling_capacity, role_adaptability, career_scalability, and investment_horizon. This softer conjunction allows partial satisfaction across transformation criteria.